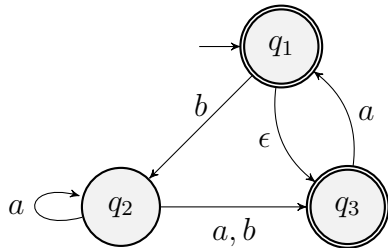


## 1 Question (12 points)



For the given NFA above with input alphabet  $\{a, b\}$ ,

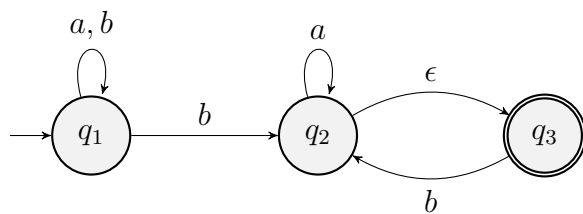
- What is (are) start state (s)?
- What is (are) final state (s)?
- What is  $Q$ ?
- Give start transition table.
- Does NFA accept string baababab? Give the states traversed by the NFA.

## 2 Question (18 points)

Given the input alphabet  $\{0, 1\}$ ,

- a. Give state diagram of DFA of  $L = \{w \mid w \text{ does not contain the substring } 110\}$
- b. Give state diagram of NFA with six states of  $L = \{w \mid w \text{ contains the substring } 10101\}$

### 3 Question (15 points)



Use the NFA to DFA conversion algorithm to convert the given NFA above to a DFA.

## 4 Question (15 points)

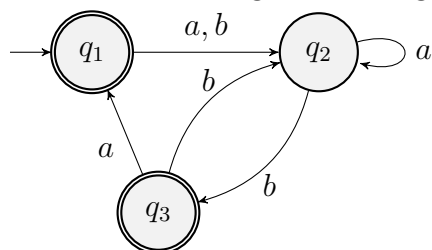
An all-NFA  $M$  is a 5-tuple  $(Q, \Sigma, \delta, q_0, F)$  that accepts  $x \in \Sigma^*$ , if every possible state that  $M$  could be in after reading input  $x$  is a state from  $F$ . Note in contrast, that an ordinary NFA accepts a string if some state among these possible states is an accept state. Prove that all-NFAs recognize the class of regular languages.

## 5 Question (20 points)

- a. Convert the following regular expression to NFA.

$$(0 \cup 1)^* 000(0 \cup 1)^*$$

- b. Convert the following NFA to regular expression.



## 6 Question (20 points)

Use the pumping lemma to prove that the following languages are nonregular.

- a.  $A = \{1^{n^3} | n \geq 0\}$
- b.  $B = \{0^n 1^n 2^n | n \geq 0\}$